



Novel supplemental guidelines for zinc, copper, and manganese in bovines, integrating net requirements, native dietary occurrence, and the boundaries of homeostatic regulation

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ABSTRACT

A long-standing objective in ruminant nutrition is to ensure an adequate supply of the key trace metals Zn, Cu, and Mn, while accounting for bioavailability and nutritional interactions, and not exceeding tolerance. In practice, however, dietary supplementation often far exceeds reference nutritional guidelines, largely as a precaution against the multiple uncertainties of trace mineral supply. Current reference recommendations are determined using fixed and highly conservative uptake coefficients, an approach that overlooks both the opportunities and the limitations of absorptive regulation, the primary homeostatic mechanism controlling trace metal balance. Although generous supplementation should mitigate deficiency risks, it also introduces the less recognized risk of exceeding tolerable availability. Recent complete balance studies demonstrate that homeostatic downregulation of trace metals can be overwhelmed at dietary levels commonly observed in practice, underscoring the need for a new framework to define supplementation. We therefore developed an approach that integrates 3 elements: (1) stochastic analysis of basal dietary supply; (2) stochastic analysis of net nutrient requirements; and (3) probabilistic assessment of the likelihood that dietary supply falls outside the lower or upper boundaries of homeostatic regulation for these metals in cattle. This last element addresses the main limitation of current dietary guidelines by considering the dynamic range of absorptive regulation: maximum uptake rates achievable during upregulation and minimum uptake rates attainable during downregulation, instead of assuming fixed absorption coefficients. As a result, this framework defines ranges of adequate dietary supply rather than a single minimum requirement. The probability distribution of nutrient supply was derived from variation in DMI, diet composition,

and trace mineral occurrence in feedstuffs, whereas the distribution of net trace metal requirements was defined by the variability of the components of a classic factorial requirement approach applied to different animal conditions. These distributions were then compared against earlier defined boundaries of homeostatic regulation. This risk-based approach supports preventive supplementation of Cu, Zn, and Mn for dairy cattle across most physiological and productive states. At the same time, it highlights the risk of exceeding tolerance thresholds under current supplementation practices, even at levels commonly considered safe. Overall, this analysis introduces novel reference guidelines to trace metal nutrition, demonstrating the opportunity to improve adequacy and reduce risk by reevaluating conventional supplementation strategies.
Key words: supplementation, trace metals, cattle, recommendation

INTRODUCTION

Dietary requirements for gross trace metal supply (e.g., INRA, 2018; NASEM, 2021) represent a critical reference to guide nutritionists in their trace metal supplementation practices. These requirements have traditionally relied on factorial assessments of net trace metal requirements, and on the assumption of a fixed true absorption efficiency. These references are used to define supplementation practices to result in dietary supplies exceeding these well-accepted gross requirement guidelines. As reported by surveys of current feeding practices, actual trace metal supply levels largely exceed gross requirements (Li et al., 2005; Castillo et al., 2013; Sinclair and Atkins, 2015; Duplessis et al., 2023), suggesting that nutritionists, in practice, apply ample safety margins above these guidelines. Indeed, a survey from Canada indicates that more than 60% of nutritionists participating in the study had intentionally formulated diets above trace element software recommendations (Duplessis et al., 2023). Such translation of gross requirement into practical application disregards

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

the opportunity and limitations of absorptive regulation, the main homeostatic mechanism of these nutrients, as reviewed by Daniel and Martín-Tereso (2025a). As trace metal supplies decrease, apparent absorption efficiency increases to match apparent retention with net requirements, until the limits of trace metal availability do not allow further improvements in efficiency. Thus, theoretically, minimal gross requirement would be calculated using the maximum apparent trace metal absorption efficiency that may be achieved under severe mineral nutrient deficient conditions. Another challenge associated with trace metal supplementation is the uncertainty of available supply and net requirements, due to insufficient information about native dietary occurrence, effects of dietary interactions, and variability in animal physiology and productivity. These uncertainties are all very valid reasons for nutritionists to formulate diets above trace element recommendations. However, there is growing concern that excessive trace metal supply may have adverse effects on the animals (e.g., the case of Cu reviewed by López-Alonso and Miranda, 2020). Hepatic Cu above the reference range is now widespread in cattle (Kendall et al., 2015; Counotte et al., 2019), and some studies have reported adverse effects on dairy cattle health and milk performance from overfeeding Cu (e.g., McCaughern, 2019). A recent study (Daniel et al., 2023) that evaluated Zn, Cu, and Mn balance relative to calving has described the overwhelming of homeostatic downregulation of trace metals at dietary levels that are not uncommon in practice. The trace metal supply at which further increase leads to increased apparent absorption above expected net requirements (Daniel and Martín-Tereso, 2025a) may therefore be a useful nutritional threshold to consider when defining trace metal supplementation.

The objectives of this study were as follows: (1) to quantify the risks of both trace metal deficiency and intakes exceeding tolerable limits across different cattle categories (dry, growing, and lactating) in nonsupplemented diets by stochastically generating representative basal trace metal supply and net trace metal requirements; (2) to evaluate how the risks change with supplemental feeding levels; and (3) to evaluate these risks in practice using a real dataset from lactating dairy cows, including animal performance data and formulated TMR from multiple commercial dairy farms.

MATERIALS AND METHODS

Simulated Individual Cattle Datasets

A total of 4 datasets of 5,000 individual records each were simulated for growing heifers, pregnant nulliparous cows, pregnant parous cows, and lactating cows. To ensure realism in the simulated datasets, we used published

ranges of variables and their intercorrelations during data generation. The simulated datasets can be found in Supplemental File S1 (see Notes).

Growing Heifer Dataset. The growing heifer dataset simulated animals from 4 to 20 mo of age (122–610 d of age). Body weight was calculated using the linear relationship between BW and month of age published in Ettema and Santos (2004; see Figure 2 from the cited manuscript), assuming a normal distribution with CV of 10%. Average daily gain was simulated based on the data presented by Bach et al. (2021), which depicts individual ADG across various age intervals. The mean ADG was assumed to decline linearly from 1.100 kg/d at 4 mo of age to 0.875 kg/d at 20 mo. Similarly, the SD of ADG was modeled to decrease linearly from 0.15 kg/d to 0.05 kg/d over the same age range. These assumptions yielded a distribution of ADG values consistent with those visually represented in Bach et al. (2021). Dry matter intake was simulated based on average pen DMI data from 6,174 measurements of Holstein heifers reported by Hoffman et al. (2008). Using the data presented in Table 1 of that study, the following predictive equation was developed: $DMI \text{ (kg/d)} = 16.123 \times (BW/1,000)^{0.6592}$, where BW is body weight in kilograms. Dry matter intake was assumed to follow a normal distribution with a CV of 10%. The consistency between simulated DMI and ADG was evaluated through the calculation of feed efficiency, defined as the ratio of ADG to DMI. This approach yielded a realistic decline in feed efficiency from 4 to 20 mo of age. Specifically, feed efficiency, defined as fractional ratio of ADG with DMI, averaged $20.7\% \pm 3.5\%$ from 122 to 152 d of age, and $8.6\% \pm 1.3\%$ from 580 to 610 d of age.

Pregnant Nulliparous and Parous Cow Datasets. Two separate datasets were created to simulate cattle in their third trimester of pregnancy, the period associated with highest gestational requirements: one for nulliparous cows and another for parous cows in their dry period. Means and SD for DMI and BW of these 2 groups of cows were retrieved from the meta-analysis of Husnain and Santos (2019) to generate normally distributed data. Dry matter intake and BW were, 9.8 ± 1.1 kg/d and 602 ± 25 kg, respectively, for nulliparous cows, and 12.2 ± 2.3 kg/d and 694 ± 50 kg, respectively, for parous cows. Based on data from another meta-analysis in lactating cows (Daniel et al., 2016), a correlation of 0.52 was assumed between DMI and BW. Mean ADG of 0.577 kg/d (Bach et al., 2021) with 0.050 kg/d as SD were assumed for nulliparous cows. For parous cows, mean ADG for first lactation, second lactation, and third and greater lactations were assumed respectively to be 0.196, 0.098, and 0.037 kg/d, as reported by Spek et al. (2022). A CV of 20% was used to simulate normally distributed ADG. Across all 5,000 simulated records, a parity distribution of 30%, 30%, and 40% was

assumed for first (1,500 records), second (1,500 records), and third and greater lactations (2,000 records). Mean and variability in calf birth weight (42.9 ± 6.0 kg; Linden et al., 2009) were also used.

Lactating Cow Datasets. Finally, a lactating cow dataset was simulated, in which 30% of the records were simulated as primiparous and 70% as multiparous. Means and SD for DMI, milk yield, and BW of these 2 groups of cows were retrieved from the meta-analysis of Husnain and Santos (2019) to generate normally distributed data. Dry matter intake, milk yield, and BW used were 16.9 ± 1.4 kg/d, 31.4 ± 3.2 kg/d, and 543 ± 26 kg, respectively, for primiparous cows, and 20.3 ± 2.9 kg/d, 37.6 ± 5.7 kg/d, and 618 ± 32 kg, respectively, for multiparous cows. For dry cows, a correlation of 0.52 was also applied between BW and DMI. Similarly, a correlation of 0.77, based on data from Daniel et al. (2016), was used between DMI and milk yield. It should be noted that the data from Husnain and Santos (2019) were mostly collected during early lactation, with a median collection period of 70 d after calving (first quartile [Q1] = 56 d, and third quartile [Q3] = 103 d). This period of collection was preferred over data representative of the mid-lactation period, as this represents the most challenging period, with highest milk production and feed efficiency.

Simulated Dietary Trace Metal Composition

To simulate variability in dietary formulation, rations were simulated for lactating cows, growing heifers, and dry cows. Diets for lactating cows and growing heifers included corn silage, grass silage, and concentrate, and straw was also considered for dry cows. For lactating cows, the inclusion rate of concentrate was simulated based on a normal distribution using the mean and SD ($47\% \pm 13\%$ of dietary DM) reported by Daniel et al. (2016). The remaining forage portion of the diet was partitioned by randomly assigning corn silage inclusion between 0% and 100% of the nonconcentrate DM, with grass silage inclusion assumed to be the remainder.

A similar approach was used for growing heifers, except that the concentrate inclusion rate was randomly drawn between 10% and 50%, consistent with data from the meta-analysis by Zanton and Heinrichs (2005). For dry cows, concentrate and straw inclusion rates were randomly sampled between 5% and 30% and 0% to 35% of dietary DM, respectively. The remaining forage component (corn and grass silage) was distributed following the same method used for lactating cow diets.

Variability in trace metal concentrations of corn silage and grass silage was simulated using the mean, standard deviation, minimum, and maximum values reported by Adams (1975), which was selected due to its large sample size. For straw, the corresponding values were derived

from wheat straw data published in Feedipedia (2025). Distributions were modeled as non-normal to reflect the typically observed short tail at low concentrations and long tail at higher concentrations.

To simulate trace metal variability in concentrate without supplemental minerals, the average trace metal concentrations from 8 feed ingredients were used: corn, distillers dried grains with solubles, barley, wheat, soybean meal, rapeseed meal, beet pulp, and wheat bran (Feedipedia, 2025). For each simulation, each raw material was assigned a random value between 0 and 25. These values were normalized to generate ingredient-specific inclusion rates, which were then used in combination with known trace metal concentrations to compute the overall trace metal content of the formulated concentrate.

Dataset Used to Evaluate Trace Metal Recommendations

To evaluate the robustness of trace metal recommendations derived from the stochastically generated datasets, a real-world dataset was used, comprising the complete dietary formulation of 1,248 TMR fed to lactating dairy cows from 508 commercial farms in southern Québec, Canada (Trouw Nutrition Canada), between July 2022 and December 2022. Detailed animal performance data, used as input for ration formulation, were available for each TMR, including the average milk yield (39.7 ± 5.3 kg/d), DMI (24.9 ± 2.2 kg/d), and BW (677 ± 49 kg). Concentrate was included in all diets at $29.9\% \pm 10.4\%$ of DM. Grass and legume silage were fed in 1,234 diets at $36.1\% \pm 14.0\%$ of DM, and corn silage was included in 992 diets at $37.6\% \pm 11.9\%$ of DM.

Although precise trace metal analyses of all forages and feedstuffs in these 1,248 diets were not available, a large regional dataset of X-ray fluorescence analyses (Trouw Nutrition Canada) was used to estimate forage trace metal concentrations. This analytical technique is recognized as accurate for trace metal determination in forages (Berzaghi et al., 2018). This supplementary dataset included samples of grass and legume silages ($n = 7,493$), grass and legume hays ($n = 2,704$), and corn silages ($n = 2,043$), all collected from the same geographical region as the dietary formulation dataset.

For each TMR, trace metal values were randomly assigned to the respective forage categories based on these X-ray fluorescence measurements. For straw and concentrate components, mineral compositions were assigned using ingredient-specific values from the feed formulation program, which were based on published literature data. For the concentrates, the contribution of the premix to trace metal supply was excluded, because the objective was to calculate the dietary trace metal content of diets before trace metal supplementation.

Calculation of Net Trace Metal Requirements

Based on the composition of carcasses of lambs and calves (Suttle, 1979), net requirements for growth were assumed, per kilogram of ADG: 24 mg Zn, 1 mg Cu, and 0.85 mg Mn. Net requirements associated with trace metals excreted in urine were calculated based on medians from Daniel et al. (2023) and expressed per day per kilogram of BW: 1.89 μ g Zn, 0.84 μ g Cu, and 0.15 μ g Mn. The same dataset was also used to estimate net trace metal requirements for lactation. The mean and SD were used to simulate a normal distribution of milk trace metal concentrations (per kilogram of milk): 4.672 ± 0.562 mg Zn, 0.083 ± 0.022 mg Cu, and 0.025 ± 0.008 mg Mn. Simulated values below minimum and above maximum values reported in Daniel et al. (2023) were discarded. Minimum values were 3.400 mg for Zn, 0.048 mg for Cu, and 0.009 for Mn, and maximum values were 5.767 mg for Zn, 0.137 mg for Cu, and 0.041 for Mn. Net trace metal requirements for lactation were calculated as milk yield multiplied by the milk trace metal concentration. As of 190 d of gestation, net requirements of the gravid uterus were also considered, and data from House and Bell (1993) were used. In this work, daily trace metal retention levels in the gravid uterus of 11.7 mg Zn, 1.569 mg Cu and 0.297 mg Mn were estimated for an expected calf birth weight of 46 kg (House and Bell, 1993). The ratio between simulated calf birth weight and 46 kg was then used to scale the daily net requirements for gestation (e.g., $42 \text{ kg}/46 \text{ kg} \times 11.7 \text{ mg Zn} = 10.7 \text{ mg Zn}$). Total daily net trace metal requirements were calculated as the sum of net requirements for growth, gestation, lactation, and urinary excretion.

Quantification of Risks of Inadequate Supply

To issue recommendations on the adequate level of supplemental trace metals for cattle, the trace metal supply of each simulated individual was compared against the lower and upper homeostatic boundaries estimated by Daniel and Martín-Tereso (2025a). The lower boundary was defined as the theoretical maximum apparent absorption efficiency achievable under constrained dietary availability. This was estimated using a dataset of published longitudinal studies in cattle, which examined variable supplies of available trace metals. In this dataset, fractional apparent absorption was calculated as the ratio of the total net requirement to total trace metal supply. For example, in Cope et al. (2009), the treatment labeled RI had an apparent Zn absorption efficiency of 11.8%, calculated as the ratio of total net requirement (168.0 mg Zn/d) to total dietary Zn supply

($62.7 \text{ mg Zn/kg DMI} \times 22.8 \text{ kg DMI} = 1,430 \text{ mg Zn/d}$). The total net Zn requirement included the following:

- Obligatory urinary Zn loss: $0.00189 \times 609 \text{ kg BW} = 1.2 \text{ mg/d}$;
- Net Zn requirement for growth: $24 \times \text{ADG}$, assumed at $0.1 \text{ kg/d} = 2.4 \text{ mg/d}$;
- Net Zn requirement for lactation: $4.672 \times 35.2 \text{ kg milk/d} = 164.5 \text{ mg/d}$.

Such calculated apparent absorption efficiency was considered plausible when no adverse effects were reported, and maximum apparent absorption efficiencies were estimated at 90% of the lowest calculated efficiency associated with adverse effects across all studies for each trace metal. These values were estimated at 17.00% for Zn and 0.45% for Mn. It should be noted that these apparent absorption coefficients are not directly comparable to true absorption coefficients of Zn (20%) and Mn (0.42%) proposed by NASEM (2021), as the former is affected negatively by the amount of endogenous metal lost through feces. For Cu, 2 scenarios were considered: one involving practical diets with supplemental antagonists (e.g., Mo, S, Fe) and one without. In this dataset, the supplemental ranges for the antagonists were as follows: Mo, 1 to 10 mg/kg (median 5 mg/kg); S, 1,500 to 2,400 mg/kg (median 1,800 mg/kg); and Fe, 500 to 800 mg/kg (median 550 mg/kg). These respective maximum apparent absorption efficiencies were 1.90% and 3.30%. It should be noted that these absorption levels reflect the efficiency expected from the basal diet, as the highest values were all observed in nonsupplemented diets. More details can be found in the review from Daniel and Martín-Tereso (2025a). As an indirect comparison, NASEM (2021) assumes true Cu absorption efficiency of 4.6% for diets low in Cu antagonists (1.0 mg of Mo and 2.0 g S/kg DM) and of 1.9% for diets high in antagonists (5.0 mg of Mo and 4.0 g S/kg DM). Thereafter, we report the proportion of diets below the lower boundary in function of supplemental trace metal levels, simulated to increase from 0 to 150 mg/kg DM, by steps of 1 mg/kg DM. Similarly, trace metal supplies of each simulated cattle were compared with proposed upper boundaries of homeostasis (Daniel and Martín-Tereso, 2025a), defined as the supply levels above which homeostasis is overwhelmed, resulting in unregulated trace metal retention. These supplies were estimated at 3.54, 0.63, and 2.36 mg/d per kilogram of BW for Zn, Cu, and Mn, respectively. The relevance of these upper boundaries was confirmed in growing bulls in which supplemental levels of Zn, Cu, and Mn were increased

Table 1. Trace metals composition of forages, concentrate and TMR fed to lactating cows in the evaluation dataset, mg/kg DM¹

Item	Median	SD	Q1	Q3	1%	99%
Grass and legume silages ² (n = 7,493)						
Zn	30.6	8.9	26.4	35.6	18.7	64.4
Cu	11.6	3.1	10.1	14.3	7.6	20.1
Mn	47.9	36.2	34.1	68.7	14.3	195.0
Grass and legume hay ² (n = 2,704)						
Zn	25.1	15.1	21.1	29.9	13.9	74.4
Cu	10.0	2.3	9.0	11.6	6.8	17.1
Mn	44.0	35.0	30.4	64.4	11.6	189.0
Corn silage ² (n = 2,043)						
Zn	24.3	6.3	21.4	28.0	15.9	47.3
Cu	7.5	1.9	6.4	8.4	4.0	13.2
Mn	17.8	10.4	13.4	24.2	6.2	56.7
Concentrate ³ (n = 1,248)						
Zn	40.9	10.2	34.1	48.1	26.3	69.9
Cu	8.5	4.8	6.6	11.4	4.7	27.6
Mn	19.4	8.4	15.2	27.2	9.8	45.5
TMR, nonsupplemented (n = 1,248)						
Zn	30.9	5.0	28.3	34.1	23.3	48.6
Cu	9.3	1.7	8.4	10.4	6.7	14.7
Mn	30.0	14.7	24.1	38.0	15.0	87.0

¹Q1, Q3 = first and third quartiles, respectively; 1% = first percentile; 99% = 99th percentile.

²Samples analyzed with X-ray fluorescence.

³Trace metal composition of the concentrate was estimated from feed table composition of individual ingredients.

exponentially (Daniel and Martín-Tereso, 2025b). In the cited study, significant increases in urinary Zn and Cu were observed when the supply exceeded the theoretical upper boundaries. These increases in urinary excretion were interpreted as indicators of overwhelmed homeostatic regulation. However, no increase in urinary Mn was reported at Mn supply of 2.92 mg/d (SUP 3 in Daniel and Martín-Tereso, 2025b) despite this level exceeding the estimated upper limit. Consequently, the upper limit used to evaluate the risk of excessive Mn supply was revised upward from 2.36 to 2.92 mg/d per kilogram of BW. This adjustment is further supported by observations suggesting that unregulated Mn retention is primarily associated with accumulation in gut tissue, rather than with systemic overload (Daniel and Martín-Tereso, 2025b). These upper boundaries were compared with the calculated trace metal supply of each individual, and the percentage of cases exceeding the upper limit was reported as a function of supplemental trace metal intakes, simulated to increase from 0 to 200 mg/kg DM by steps of 1 mg/kg DM.

RESULTS

The dietary concentrations of Zn, Cu, and Mn were similar among basal diets simulated for each different category of cattle, with mean ($\pm$ SD) values of 28.0 ($\pm$ 10.1), 8.1 ($\pm$ 3.3), and 41.0 ($\pm$ 17.8) mg/kg DM, respectively, for prepartum cows; 36.9 ($\pm$ 8.9), 9.0 ($\pm$ 2.7), and 45.6 ($\pm$ 15.2) mg/kg DM for lactating

cows; and 33.0 ($\pm$ 10.9), 9.0 ($\pm$ 3.6), and 44.3 ($\pm$ 18.7) mg/kg DM for growing heifers. Thus, differences in trace metal supplies between categories of cattle were mainly driven by difference in DMI: 9.8 ($\pm$ 1.1) kg/d for prepartum nulliparous, 12.2 ($\pm$ 2.3) kg/d for prepartum parous, 19.2 ($\pm$ 2.9) kg/d for lactating cows, and 7.9 ($\pm$ 1.8) kg/d for growing heifers. Nevertheless, within each category of cattle, variability in dietary concentration had important influences on total trace metal intake. For example, in lactating cows, at the average DMI, increasing the dietary Zn, Cu, and Mn contents of the lactating cow diet by 2 SD would raise total Zn, Cu, and Mn intake by 342, 104, and 584 mg/d, respectively. By comparison, increasing DMI of lactating cow by 2 SD at the average dietary Zn, Cu, and Mn contents would result in smaller increases in Zn, Cu, and Mn intake, estimated at 214, 52, and 264 mg/d, respectively. The simulated mean ($\pm$ SD) Zn, Cu, and Mn intakes were 292 ($\pm$ 104), 87 ($\pm$ 34), and 438 mg/d ($\pm$ 181), respectively, for prepartum nulliparous; 355 ($\pm$ 140), 106 ($\pm$ 47), and 538 mg/d ($\pm$ 245) for prepartum parous; 730 ($\pm$ 204), 182 ($\pm$ 60), and 929 mg/d ($\pm$ 328) for lactating cows; and 274 ($\pm$ 108), 77 ($\pm$ 35), and 379 mg/d ($\pm$ 175) for growing heifers. The sum of net Zn, Cu, and Mn requirements for the simulated dataset were 25.9 ($\pm$ 1.9), 2.5 ($\pm$ 0.2), and 0.86 mg/d ($\pm$ 0.06), respectively, for prepartum nulliparous; 14.7 ($\pm$ 2.2), 2.1 ($\pm$ 0.2), and 0.47 mg/d ($\pm$ 0.07) for prepartum parous; 169 ($\pm$ 32), 3.6 ($\pm$ 0.8), and 1.06 mg/d ($\pm$ 0.28) for lactating cows; and 24.4 ($\pm$ 2.9), 1.3 ($\pm$ 0.1), and 0.89 mg/d ($\pm$ 0.10) for growing heifers.

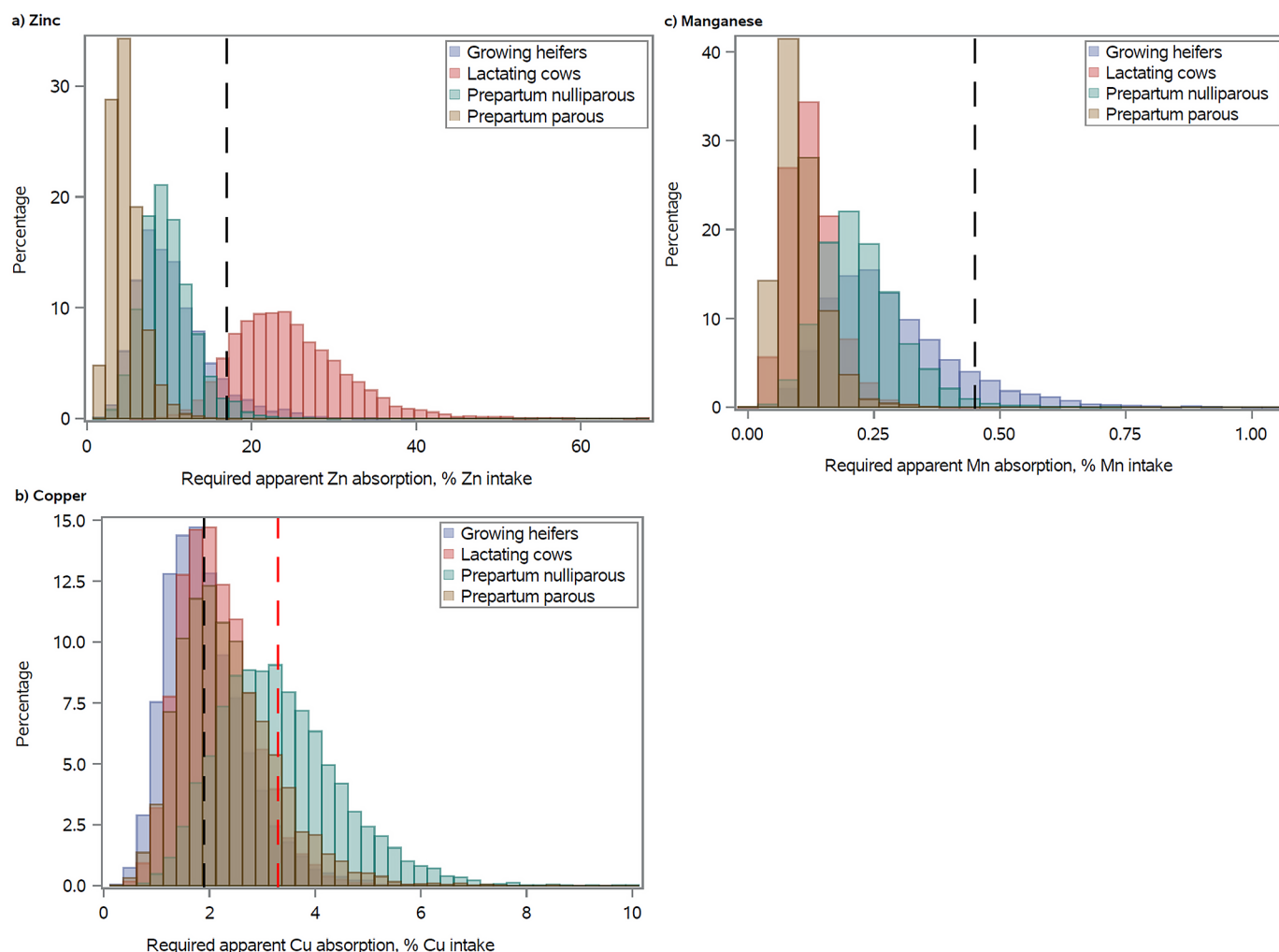


Figure 1. Histograms of the required apparent absorption efficiency for (a) Zn, (b) Cu, and (c) Mn to balance dietary supply with net requirements. The vertical black dashed lines indicate the theoretical maximum achievable apparent absorption efficiency. For Cu, black and red lines represent the maximum absorption efficiency with and without dietary antagonists, respectively.

Expressed as coefficient of variation, the variability in trace metal supply within each cattle category was considerably greater than the variability in net requirements. For nonlactating cattle, the supply variability ranges from 36% to 46% of the mean, compared with 7% to 15% for their net requirements. In lactating cows, supply variability ranged from 28% to 35%, and net requirement variability ranged from 19% to 26%. This underscores the critical role of trace metal homeostasis in buffering against fluctuations in dietary supply.

The results of the trace metal analysis in feedstuffs and complete diets fed to lactating cows, based on a real-world dataset used to evaluate the soundness of the derived recommendations, are presented in Table 1. As expected from variance reduction by mixing, the coefficients of variation were higher for individual feedstuff categories than for nonsupplemented TMR.

For example, the CV for Mn in corn silage and in grass and legume silages were 58% and 76%, respectively, whereas the CV for Mn in the TMR was only 49%. In comparison, the variability in Zn and Cu concentrations within TMR was lower, with CV of 16% and 18%, respectively. The dataset was characterized by higher average values for DMI (25.0 ± 2.4 kg/d) of lactating cows compared with those simulated, which explains the higher supply of Zn (792 ± 146 mg/d) and Cu (237 ± 47 mg/d). Despite the higher DMI, Mn supply was lower (843 ± 395 mg/d), due to the lower Mn concentration in the TMR (30.0 ± 14.7 mg/kg DM; Table 1). Consistent with the higher milk yield (39.8 ± 5.2 kg/d) and BW (676 ± 50 kg), net requirements for Zn, Cu, and Mn were higher than those simulated, with mean values of $187 (\pm 31)$, $4.1 (\pm 0.9)$, and 1.20 mg/d (± 0.29), respectively.

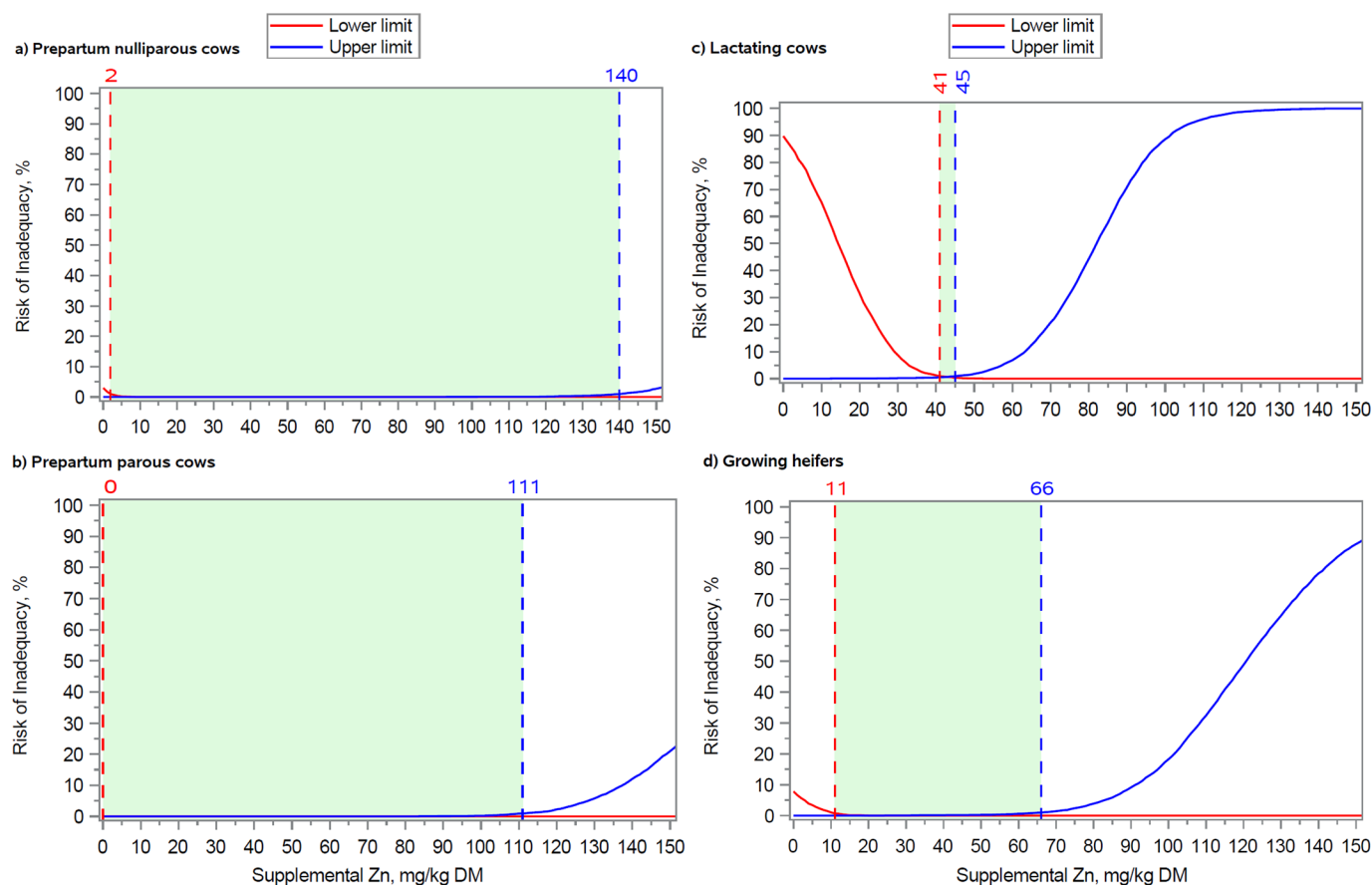


Figure 2. Risk of Zn supply inadequacy as a function of supplemental Zn in (a) prepartum nulliparous cows, (b) prepartum parous cows, (c) lactating cows, and (d) growing heifers. Red vertical dashed line represents the lower supplemental Zn level at which the risk of undersupply is below 1%. Blue vertical dashed line represents the upper supplemental Zn level at which the risk of oversupply is below 1%. The shaded green area represents the range of supplemental Zn that minimizes both risks below 1%.

Risk of Insufficient Trace Metal Supply with No Supplemental Trace Metals

Figure 1 illustrates the theoretical apparent trace metal absorption efficiencies that would be required to meet net trace metals requirements for cattle fed diets not supplemented with trace metals. For Zn, lactating cows were by far at the greatest risk, as 88.7% of cases had required apparent Zn absorption efficiency that exceeded the theoretical maximum of 17.0% to meet their estimated net needs. In contrast, the risks of undersupply appeared low for prepartum nulliparous heifers, prepartum parous cows, and growing heifers, with only 2.9%, 0.0%, and 7.8% of animals, respectively, having dietary Zn supply considered insufficient.

The risk of insufficient Cu supply was evaluated using 2 threshold values: one representing conditions with elevated Cu antagonists (Mo, S, and Fe) and the other assuming the absence of such antagonists. Under high-antagonist conditions, the estimated risks of Cu undersupply

were substantial: 91.0% for prepartum nulliparous heifers, 64.5% for prepartum parous cows, 60.0% for lactating cows, and 45.6% for growing heifers. When dietary Cu antagonist levels were low, these risks decreased markedly to 46.3%, 14.9%, 6.4%, and 5.8%, respectively.

The risk of Mn underfeeding was estimated to be very low for most cattle categories. No cases of insufficient Mn supply were identified among prepartum parous cows or lactating cows, and only 1.2% of prepartum nulliparous heifers had Mn intake considered inadequate. In contrast, a notable proportion of growing heifers (10.4%) were at risk of Mn undersupply based on basal diet alone. The risk was particularly elevated among younger, more feed-efficient animals: 27.5% of first-year heifers (122 to 365 d of age) were at risk, compared with only 3.2% of second-year heifers (366 to 610 d of age; results not shown).

In the evaluation dataset, the risk of inadequate trace metal supply for lactating cows was similar across trace metals. Specifically, 94.2% of cases fell below the lower

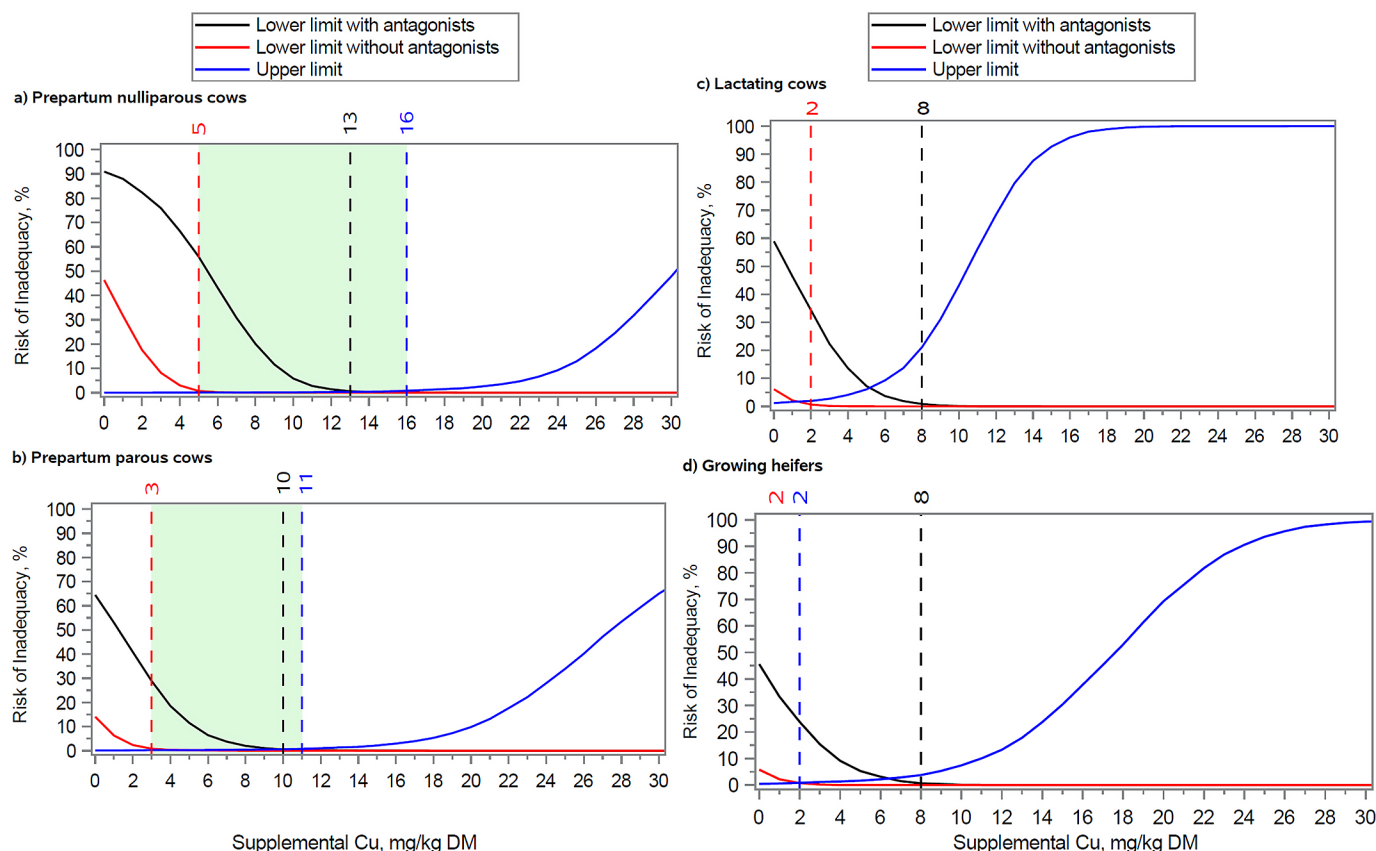


Figure 3. Risk of Cu supply inadequacy as a function of supplemental Cu in (a) prepartum nulliparous cows, (b) prepartum parous cows, (c) lactating cows, and (d) growing heifers. Black and red vertical dashed lines represent the lower supplemental Cu level at which the risk of undersupply is below 1% with and without dietary antagonists, respectively. Blue vertical dashed line represents the upper supplemental Cu level at which the risk of oversupply is below 1%. The shaded green area, when present, represents the range of supplemental Cu that minimizes both risks below 1%.

limit for Zn, 6.0% of cases (or 33.7% in the presence of Cu antagonists) fell below the lower limit for Cu, and 0.1% of cases fell below the lower limit for Mn.

Consequences of Supplemental Trace Metal Levels on Risks for Inadequate Supply

The effects of supplemental trace metal levels on the percentage of cattle falling below the lower limit or above the upper limit are shown for Zn in Figure 2, Cu in Figure 3, and Mn in Figure 4. Table 2 also indicates the minimum and maximum supplemental trace metal levels to maintain risks of undersupply and oversupply below 1% and 5%. Figure 2a and 2b and Table 2 indicate that the risk of inadequate Zn supply was kept low over a wide range of supplemental Zn for prepartum cattle. In contrast, for lactating cows, supplemental Zn levels that keep the risk of inadequate Zn supply below 1% were relatively narrow, from 41 to 45 mg/kg DM (Figure 2c, Table 2). The challenge of simultaneously minimizing risks of Cu underfeeding and overfeeding

for lactating cows is depicted in Figure 3c and Table 2. Even when no supplemental Cu is fed, 1.2% of lactating cows are already at risk of exceeding their Cu tolerance threshold. Supplementing 10 mg Cu/kg DM reduced the risk of undersupply from 6.4% (or 60.0% when all diets were assumed to contain high levels of Cu antagonists) to 0% but increased the risk of excess Cu from 1.2% to 43.2%. No supplemental Cu level was able to reduce the risk of inadequate supply below 1%. In contrast to lactating cows, supplementing 10 mg Cu/kg DM to prepartum cattle did not increase the risk of exceeding the tolerance threshold (Figure 3a, 3b). These results highlight the importance of considering the balance between total Cu supply and net Cu demands when evaluating the effects of a feeding strategy. Similarly, the risk of exceeding the upper Mn supply threshold was highest for lactating cows, intermediate for growing heifers, and lowest for prepartum cattle (Table 2 and Figure 4).

In the evaluating dataset, containing 1,248 lactating cow diets fed from commercial farms in the province of

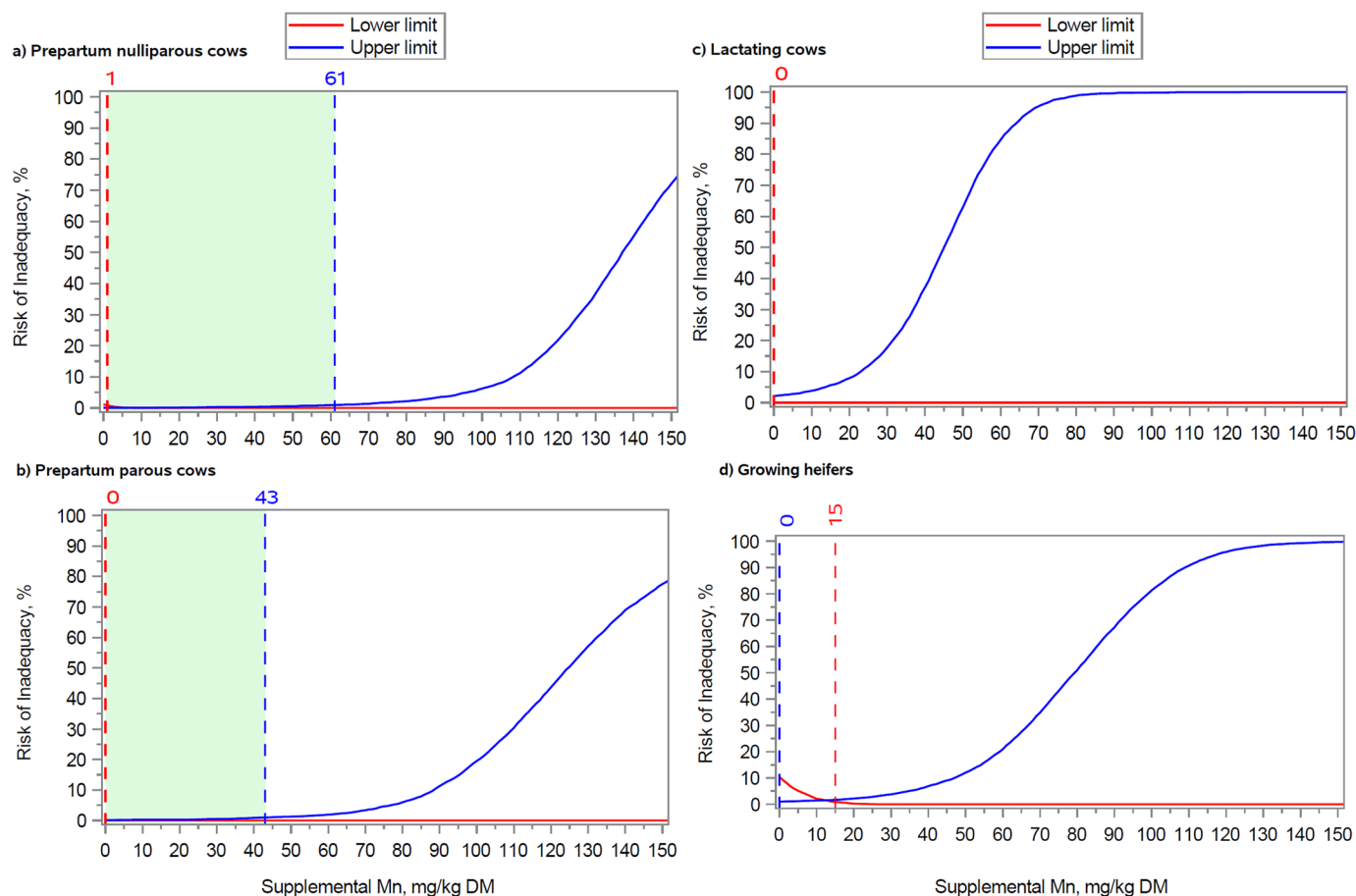


Figure 4. Risk of Mn supply inadequacy as a function of supplemental Mn in (a) prepartum nulliparous cows, (b) prepartum parous cows, (c) lactating cows, and (d) growing heifers. Red vertical dashed line represents the lower supplemental Mn level at which the risk of undersupply is below 1%. Blue vertical dashed line represents the upper supplemental Mn level at which the risk of oversupply is below 1%. The shaded green area, when present, represents the range of supplemental Mn that minimizes both risks below 1%.

Québec, Canada, the ranges of supplemental levels that minimized the number of cases outside recommended boundaries were remarkably similar to those calculated using the simulated dataset (Table 3). Interestingly, the ranges of supplemental Zn, Cu, and Mn that minimized the risk of undersupply below 1% and oversupply below 5% in the simulated lactating cow dataset fully overlapped with the corresponding ranges calculated from this regional dataset.

DISCUSSION

The proposed method provides a novel approach to defining trace metal supplementation guidelines in cattle, offering several key improvements over traditional methods.

The first is the explicit acknowledgment of the regulatory role played by apparent trace metal absorption in cattle (Daniel and Martín-Tereso, 2025a). In contrast to classical factorial models (e.g., NASEM, 2021),

which assume a fixed true absorption coefficient to convert net into gross trace metal requirements, the current approach estimates the apparent absorption needed to meet net requirements. Consequently, the apparent absorption values can vary substantially. The feasibility of achieving these absorption levels is then evaluated against potential maximum uptake rates previously derived from published literature (Daniel and Martín-Tereso, 2025a).

The second novelty is the consideration of upper supply values associated with an increased risk of unregulated trace metal retention, reflecting the point at which cattle may no longer be able to downregulate uptake (Daniel et al., 2023; Daniel and Martín-Tereso, 2025a,b). Rather than defining a single minimum requirement to exceed, this approach allows for defining dietary supply ranges within which cattle can rely on homeostasis, thereby operating in a supply range for optimal performance and health. By providing a recommended range of dietary

Table 2. Minimum and maximum supplemental zinc, copper, and manganese levels (mg/kg DM) to maintain risks of undersupply and oversupply within acceptable limits for different cattle category¹

Item	Lower 1%	Lower 5%	Upper 5%	Upper 1%
Zinc				
Prepartum nulliparous	2	0	157	140
Prepartum parous	0	0	128	111
Lactating cows	41	34	57	45
Growing heifers	11	3	82	66
Copper				
Prepartum nulliparous	5 (13)	4 (11)	22	16
Prepartum parous	3 (10)	2 (7)	17	11
Lactating cows	2 (8)	1 (6)	4	Unfeasible ²
Growing heifers	2 (8)	1 (6)	8	2
Manganese				
Prepartum nulliparous	1	0	95	61
Prepartum parous	0	0	76	43
Lactating cows	0	0	13	Unfeasible ³
Growing heifers	15	6	34	0

¹Minimum supplemental trace metal level (mg/kg DM) required to limit the proportion of cows with dietary supply below the lower boundary to <1% or <5%, and maximum supplemental level (mg/kg DM) required to limit the proportion of cows exceeding the upper boundary to <1% or <5%. For each cattle category, 5,000 individuals were simulated. Numbers in parentheses are minimum level of supplemental Cu that minimize the risk of undersupply below 1% and 5%, respectively, under scenario of high dietary antagonists.

²Without supplemental Cu, 1.2% of cases were above the upper boundary. Thus, no scenario minimizes the risk below 1%.

³Without supplemental Mn, 2.1% of cases were above the upper boundary. Thus, no scenario minimizes the risk below 1%.

supplies, the guidelines not only emphasize the risk associated with possible deficiency but also the risks linked to exceeding the tolerable threshold. This emphasis is important, as surveys reveal that feeding practices largely exceed gross requirements (Li et al., 2005; Castillo et al., 2013; Sinclair and Atkins, 2015; Duplessis et al., 2023), explaining for the most part the high level of hepatic Cu observed in dairy cattle (Kendall et al., 2015; Counotte et al., 2019; Strickland et al., 2019), which may impair milk performance and reproduction (McCaughern, 2019; McCaughern et al., 2024).

A third improvement proposed by our approach is that we based the recommendations on the plausible range of variation in trace metal supplies and net requirements, thereby representing the diversity of cows and diets under indoor feeding conditions. This contrasts with traditional approaches, which typically rely on average cow characteristics (milk yield, BW, and DMI) to define gross trace mineral requirements and supply. By explicitly describing and quantifying these sources of variability, we create several opportunities. First, it allows us to assess, for any supplementation practice, the actual risk of mismatch between trace metal supply and recommended range. Second, it provides clearer insight into the uncertainty arising from both basal dietary trace metal concentrations and animal variability. As a result, we can propose ranges of supplemental trace metal supply that minimize the dual risks of underfeeding, with potential risk of deficiency, and overfeeding, exceeding tolerable thresholds.

This framework also enables direct comparison between supplementation practices of nutritionists and the recommended ranges established for each category of cattle. The direct consideration of basal trace metal supply when issuing supplementation guidelines is needed, as it is not systematically considered by nutritionists (Duplessis et al., 2023), which can partly explain the use of a high safety margin. Our analysis confirms the important contribution of the basal diet to meeting trace metal requirements. Except for Zn during lactation and Cu before first calving, our results suggest that in most dairy cattle diets (at least two-thirds), requirements can be met without additional trace metal supplementation. In this context, preventive supplementation functions much like an insurance policy: its purpose is not to cover average situations but to hedge the risk of deficiency in uncertain nutritional contexts where the basal supply may prove inadequate, whether due to specific animal performance, dietary composition, or their interaction.

The recommended range of supplemental trace metal is expressed as a concentration (mg/kg DMI) rather than as daily intake. Although this choice may initially appear to simplify the problem, it more appropriately reflects the way supplementation is typically applied in practice. Indeed, when cows are fed a TMR, one decides on the inclusion level of supplemental trace metal per kilogram of feed offered, whereas the actual daily intake of supplemental trace metals is ultimately driven by voluntary feed intake. The role of our risk-based

Table 3. Minimum and maximum supplemental zinc, copper, and manganese levels (mg/kg DM) to maintain risks of undersupply and oversupply within acceptable limits for lactating cows of the evaluation dataset¹

Item	Lower 1%	Lower 5%	Upper 5%	Upper 1%
Zinc	30	25	57	48
Copper	0 (5)	0 (3)	4	1
Manganese	0	0	19	Unfeasible ²

¹Minimum supplemental trace metal level (mg/kg DM) required to limit the proportion of cows with dietary supply below the lower boundary to <1% or <5%, and maximum supplemental level (mg/kg DM) required to limit the proportion of cows exceeding the upper boundary to <1% or <5%. Numbers in parentheses are minimum level of supplemental Cu that minimize the risk of undersupply below 1% and 5%, respectively, under scenario of high dietary antagonists.

²Without supplemental Mn, 2.4% of cases were above upper boundary. Thus, no scenario minimizes the risk below 1%.

analysis is therefore to evaluate the implication of adding a certain amount of trace metal in milligrams per kilogram of DM across a wide range of basal dietary conditions and a wide range of cattle. This approach clearly demonstrates that adding 10 mg of supplemental Cu per kilogram of DM has markedly different effects on the risk of exceeding the tolerable threshold across cattle categories. The estimated risks increased from 0.4% in prepartum cattle, to 7.4% in growing cattle, and up to 43.2% in lactating cows. This pattern is mainly explained by a greater imbalance between dietary trace metal supply and net requirements, largely driven by differences in voluntary DMI. These results seem consistent with reports that a higher fraction of dairy cows, compared with bulls and beef cows, had hepatic Cu above reference range (Kendall et al., 2015). The influence of lactation on hepatic Cu has also been described in a study that evaluated hepatic Cu concentration in Holstein heifers receiving either 15 or 30 mg Cu/kg DM (McCaughern, 2019). Hepatic Cu concentration first decreased between 7 mo of age (643 and 798 mg/kg DM, respectively) and 6 wk before expected calving (128 and 293 mg/kg DM, respectively), but increased thereafter, as analyzed 14 wk after parturition (424 and 607 mg/kg DM, respectively).

As illustrated in Tables 2 and 3, it is not always possible to define a single inclusion level that simultaneously minimizes the risks of both under- and oversupplementation across all conditions. For example, assuming all lactating cow diets contain high levels of Cu antagonists (e.g., Mo), supplementation with 8 mg Cu/kg DM would be required to reduce the risk of underfeeding to below 1%. However, this practice would raise the risk of overfeeding from 1.2% to 21.0%. This demonstrates that a uniform Cu inclusion level cannot simultaneously control both risks below 1% and underscores the importance of dietary analysis. In contrast, for growing heifers, any Zn supplementation between 11 and 66 mg/kg DM would be expected to maintain both risks below 1% regardless of growth rate, body size, or diets fed.

It should be noted that in the current approach, the potential contribution of body trace metal reserves to compensate for inadequate dietary supply was not considered. Although this omission may be physiologically justified for Zn and Mn, the liver represents a substantial mobilizable reserve for Cu in cattle. For example, a prepartum cow with high net Cu requirements would need ~3.0 mg Cu/d to cover obligatory urinary losses, growth, and gestational needs. If, 90 d before calving, hepatic Cu concentration is 300 mg/kg DM and liver size is 2.0 kg DM, fully meeting these requirements from hepatic reserves would reduce hepatic Cu concentration to 150 mg/kg DM at parturition. Such a level remains comfortably above the critical threshold (~15 to 20 mg/kg DM; Daniel and Martín-Tereso, 2025a) associated with a decline in blood Cu. Moreover, because dams appear to prioritize Cu transfer to the calf (Engel et al., 1964), such a situation would not be expected to compromise calf health.

The potential contribution of body Zn and Mn to compensate for inadequate dietary intake remains uncertain. Studies in rats have shown that when a severely Zn-deficient diet (2 mg/kg DM) was fed for 12 d, ~21% of bone Zn was mobilized, and this deficit was replenished within 2 d upon refeeding a Zn-adequate diet. In a 458-kg cow, total bone Zn has been reported to be 2,289 mg (Neathery et al., 1973). If a similar proportion of bone Zn were mobilizable in cows, ~480 mg of Zn could potentially be released. This amount is only modestly higher than the daily Zn requirement for a cow producing 30 kg of milk, which is about 135 mg/d. These data suggest that the capacity to compensate for dietary Zn deficiency via bone stores is limited and less significant than for Cu. For Mn, the compensatory role of body stores appears to be even more restricted. In lambs fed a Mn-deficient diet (0.8 mg/kg DM) for 22 wk, the concentration of Mn in bone, the largest Mn pool in the body, remained unchanged, despite clinical signs of Mn deficiency appearing after 7 wk of feeding (Lassiter and Morton, 1968).

In contrast to Cu, the available literature did not support constraining the maximal apparent absorption efficiency

of Zn and Mn in the presence of high dietary antagonist concentrations (Daniel and Martín-Tereso, 2025a). Nonetheless, a simple sensitivity analysis was conducted to evaluate the effects of under- or overestimating maximal Zn absorption by $\pm 20\%$. Assuming maximum absorption efficiencies of 13.6%, 17.0% (baseline), and 20.4%, the supplemental Zn required to maintain deficiency risk below 1% in lactating cows ranged from 30 to 57 mg/kg DM (30, 41, and 57 mg/kg DM, respectively). These results highlight the uncertainty surrounding the supplemental Zn level required to minimize the risk of under-supply. However, to mitigate the risk, precautions were taken when estimating the maximum apparent absorption efficiency. First, the lowest efficiency associated with adverse effects, rather than means of efficiency across all treatments with reported adverse effects, was used to define this maximum. Second, a safety factor of 10% was applied. Finally, it is not possible to distinguish true prevention of deficiency from performance improvement resulting from trace metal supplementation beyond the net requirement. For example, Wang et al. (2021) reported a linear increase in milk production when dietary Zn concentration increased from 31 to 61 mg/kg DM. Because the observed improvements were relatively small and linear, this response may reflect a feed-additive-like effect rather than correction of a deficiency, which would be expected to elicit a more rapid and pronounced improvement in productivity. Consequently, our approach to estimating the lower boundary, which may partially include performance-enhancing effects, should be considered conservative.

Whereas the primary focus of the work was on quantifying the possible implication of dietary level, in practice, various supplementation forms of trace metals are fed to cattle. The potential effect of trace metal source was not explicitly considered in the guideline. However, it can be speculated that mineral form may influence both the lower and upper thresholds of the recommended range. It would be of interest for future studies to evaluate these hypotheses under conditions relevant to cattle nutrition.

CONCLUSIONS

The risk-based approach presented in the current paper offers a novel perspective to trace metal supplementation, pointing at an opportunity to improve nutritional adequacy by reconsidering supplementation practices. This approach confirms the value of preventive supplementation of Zn, Cu, and Mn to dairy cattle in most representative physiological states. It also illustrates the importance of native dietary supply to fulfill requirements, as, except for Zn in lactation and Cu during the dry period, native trace metal supplies of Zn, Cu, and Mn were estimated to cover the requirements of two-thirds of

cattle. Risk of exceeding trace metal tolerance boundaries with common levels of supplementation under common conditions were also demonstrated. Evaluating the effect of the proposed recommendation for trace metal levels against current commercial practice on cattle health and performance should validate the value of our approach.

NOTES

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Nonstandard abbreviations used: Q1 = first quartile; Q3 = third quartile.

REFERENCES

- Adams, R. S. 1975. Variability in mineral and trace element content of dairy cattle feeds. *J. Dairy Sci.* 58:1538–1548. [https://doi.org/10.3168/jds.S0022-0302\(75\)84750-3](https://doi.org/10.3168/jds.S0022-0302(75)84750-3).
- Bach, A., J. Ahedo, and A. Kertz. 2021. Invited review: Advances in efficiency of growing dairy replacements. *Appl. Anim. Sci.* 37:404–417. <https://doi.org/10.15232/aas.2021-02164>.
- Berzaghi, P., A. Lotto, M. Mancinelli, and F. Benozzo. 2018. Technical note: Rapid mineral determination in forages by X-ray fluorescence. *J. Dairy Sci.* 101:9967–9970. <https://doi.org/10.3168/jds.2018-14740>.
- Castillo, A. R., N. R. St-Pierre, N. Silva del Rio, and W. P. Weiss. 2013. Mineral concentrations in diets, water, and milk and their value in estimating on-farm excretion of manure minerals in lactating dairy cows. *J. Dairy Sci.* 96:3388–3398. <https://doi.org/10.3168/jds.2012-6121>.
- Cope, C. M., A. M. Mackenzie, D. Wilde, and L. A. Sinclair. 2009. Effects of level and form of dietary zinc on dairy cow performance and health. *J. Dairy Sci.* 92:2128–2135. <https://doi.org/10.3168/jds.2008-1232>.
- Counotte, G., M. Holzhauer, S. Carp-van Dijken, J. Muskens, and D. Van der Merwe. 2019. Levels of trace elements and potential toxic elements in bovine livers: A trend analysis from 2007 to 2018. *PLoS One* 14:e0214584. <https://doi.org/10.1371/journal.pone.0214584>.
- Daniel, J. B., D. Brugger, S. van der Drift, D. van der Merwe, N. Kendall, W. Windisch, J. Doelman, and J. Martín-Tereso. 2023. Zinc, copper, and manganese homeostasis and potential trace metal accumulation in dairy cows: Longitudinal study from late lactation to subsequent mid-lactation. *J. Nutr.* 153:1008–1018. <https://doi.org/10.1016/j.tjnut.2023.02.022>.
- Daniel, J. B., N. C. Friggens, P. Chapoutot, H. Van Laar, and D. Sauvant. 2016. Milk yield and milk composition responses to change in predicted net energy and metabolizable protein: A meta-analysis. *Animal* 10:1975–1985. <https://doi.org/10.1017/S1751731116001245>.
- Daniel, J. B., and J. Martín-Tereso. 2025a. Review: Homeostatic boundaries to dietary Zn, Cu and Mn supply in cattle. *Animal* 19:101532. <https://doi.org/10.1016/j.animal.2025.101532>.

- Daniel, J. B., and J. Martín-Tereso. 2025b. Effects of supplemental levels of zinc, copper, and manganese on apparent absorption and tissue retention in growing bulls. *J. Dairy Sci.* 108:8410–8427. <https://doi.org/10.3168/jds.2025-26774>.
- Drenth, P. J. D. 2012. The European code of conduct for research integrity. Pages 161–168 in *Promoting Research Integrity in a Global Environment*. T. Mayer and N. Steneck, ed. World Scientific Publishing Co. Pte. Ltd., Singapore City, Singapore.
- Duplessis, M., T. C. Wright, and M. Bejaei. 2023. A survey of Canadian dairy nutritionists to assess current trace element formulation practices. *J. Dairy Sci.* 106:4030–4041. <https://doi.org/10.3168/jds.2022-22943>.
- Engel, R. W., W. A. Hardison, R. P. Miller, N. O. Price, and J. T. Huber. 1964. Effect of copper intake on concentration in body tissue and on growth, reproduction and production in dairy cattle. *J. Anim. Sci.* 23:1160–1163. <https://doi.org/10.2527/jas1964.2341160x>.
- Ettema, J. F., and J. E. P. Santos. 2004. Impact of age at calving on lactation, reproduction, health, and income in first-parity Holsteins on commercial farms. *J. Dairy Sci.* 87:2730–2742. [https://doi.org/10.3168/jds.S0022-0302\(04\)73400-1](https://doi.org/10.3168/jds.S0022-0302(04)73400-1).
- Feedipedia. 2025. Feedipedia: A programme by INRAE, CIRAD, AFZ and FAO. www.feedipedia.org.
- Hoffman, P. C., K. A. Weigel, and R. M. Wernberg. 2008. Evaluation of equations to predict dry matter intake of dairy heifers. *J. Dairy Sci.* 91:3699–3709. <https://doi.org/10.3168/jds.2007-0644>.
- House, W. A., and A. W. Bell. 1993. Mineral accretion in the fetus and adnexa during late gestation in Holstein cows. *J. Dairy Sci.* 76:2999–3010. [https://doi.org/10.3168/jds.S0022-0302\(93\)77639-0](https://doi.org/10.3168/jds.S0022-0302(93)77639-0).
- Husnain, A., and J. E. P. Santos. 2019. Meta-analysis of the effects of prepartum dietary protein on performance of dairy cows. *J. Dairy Sci.* 102:9791–9813. <https://doi.org/10.3168/jds.2018-16043>.
- INRA. 2018. INRA Feeding System for Ruminants. Wageningen Academic Publishers, Wageningen, the Netherlands.
- Kendall, N. R., H. R. Holmes-Pavord, P. A. Bone, E. L. Ander, and S. D. Young. 2015. Liver copper concentrations in cull cattle in the UK: Are cattle being copper loaded? *Vet. Rec.* 177:493. <https://doi.org/10.1136/vr.103078>.
- Lassiter, J. W., and J. D. Morton. 1968. Effects of a low manganese diet on certain ovine characteristics. *J. Anim. Sci.* 27:776–779. <https://doi.org/10.2527/jas1968.273776x>.
- Li, Y., D. F. McCrory, J. M. Powell, H. Saam, and D. Jackson-Smith. 2005. A survey of selected heavy metal concentrations in Wisconsin dairy feeds. *J. Dairy Sci.* 88:2911–2922. [https://doi.org/10.3168/jds.S0022-0302\(05\)72972-6](https://doi.org/10.3168/jds.S0022-0302(05)72972-6).
- Linden, T. C., R. C. Bicalho, and D. V. Nydam. 2009. Calf birth weight and its association with calf and cow survivability, disease incidence, reproductive performance, and milk production. *J. Dairy Sci.* 92:2580–2588. <https://doi.org/10.3168/jds.2008-1603>.
- López-Alonso, M., and M. Miranda. 2020. Copper supplementation, a challenge in cattle. *Animals (Basel)* 10:1890. <https://doi.org/10.3390/ani10101890>.
- McCaughern, J. H. 2019. Optimising the Health and Performance of Dairy Cows Through Improved Copper Nutrition. PhD thesis. Department of Animal Production, Harper Adams University, Edgmond, UK.
- McCaughern, J. H., A. M. Mackenzie, E. C. L. Bleach, and L. A. Sinclair. 2024. Overfeeding copper during rearing affects the liver function and fertility of replacement dairy heifers. *Vet. Rec.* 195:e4397. <https://doi.org/10.1002/vetr.4397>.
- NASEM. (National Academies of Sciences, Engineering, and Medicine). 2021. Nutrient Requirements of Dairy Cattle. 8th rev. ed. National Academies Press, Washington, DC, USA. <https://doi.org/https://doi.org/10.17226/25806>.
- Neathery, M. W., W. P. Miller, D. M. Blackmon, R. P. Gentry, and J. B. Jones. 1973. Absorption and tissue zinc content in lactating dairy cows as affected by low dietary zinc. *J. Anim. Sci.* 37:848–852. <https://doi.org/10.2527/jas1973.373848x>.
- Sinclair, L. A., and N. E. Atkins. 2015. Intake of selected minerals on commercial dairy herds in central and northern England in comparison with requirements. *J. Agric. Sci.* 153:743–752. <https://doi.org/10.1017/S0021859614001026>.
- Spek, J. W., D. Van Wesemael, and J. De Boever. 2022. Body weight of Holstein Friesian cows. In *CVB Documentation Report*, vol. 78. CVB, Wageningen, the Netherlands.
- Strickland, J. M., T. H. Herdt, D. G. Sledge, and J. P. Buchweitz. 2019. Short communication: Survey of hepatic copper concentrations in Midwest dairy cows. *J. Dairy Sci.* 102:4209–4214. <https://doi.org/10.3168/jds.2018-15566>.
- Suttle, N. F. 1979. Copper, iron, manganese and zinc concentrations in the carcasses of lambs and calves and the relationship to trace element requirements for growth. *Br. J. Nutr.* 42:89–96. <https://doi.org/10.1079/BJN19790092>.
- Windisch, W. 2001. Homeostatic reactions of quantitative Zn metabolism on deficiency and subsequent repletion with Zn in ⁶⁵Zn-labeled adult rats. *Trace Elem. Electrolytes* 18:122–128.
- Zanton, G. I., and A. J. Heinrichs. 2005. Meta-analysis to assess effect of prepubertal average daily gain of Holstein heifers on first lactation production. *J. Dairy Sci.* 88:3860–3867. [https://doi.org/10.3168/jds.S0022-0302\(05\)73071-X](https://doi.org/10.3168/jds.S0022-0302(05)73071-X).

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